

Embedded Electrical Integration for Startup, Teleoperation, and Future Autonomy in Unmanned Ground Vehicles (UGVs)



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Introduction

Autonomous Ground Vehicles (AGVs) are robotic platforms designed to operate with limited human intervention. This project focuses on studying the embedded and electrical systems integration of the Robotnik Guardian, an Unmanned Ground Vehicle (UGV) intended for autonomous applications in search and rescue and combat search and rescue scenarios. This project represents a study of embedded system diagnostic skills, an enhanced system-level understanding of control systems, and advances the overall readiness of the platform for future expansion into autonomous applications.

Purpose

This study aims to understand and improve the Guardian's current functionality by establishing safe startup behavior and implementing reliable motion control, understand the robot's supervised electrical architecture, validating safety-critical components such as fuses and relays, and deploying a Robot Operating System (ROS)-based motion control system to ensure consistent startup and teleoperation.

Goals

- Headless Boot Startup
- Autonomous Navigation Planning
- Sensor Integration & Obstacle Detection
- AI Integration

Methodology

Power Pathway Validation:

- Traces and documented the startup logic and supervised supply routing. The LED status indicators were observed on the HDC2472 to confirm system health.

Electrical Systems Diagnosis:

- Conducted continuity testing of safety-critical components (relays, fuses, wiring paths) and verified voltage paths from battery to CNS, relays, and motion control board.

Real-time Data Logging:

- Conducted data logging on CPU for communication with notepad. This is used to capture what is being executed.

Results

The results from all the methods mentioned were successful due to the knowledge gained about the tank. The tank will not move unless there's a monitor plugged in to the CPU. A short-term fix is plugging in a HDMI 4K UHD, Virtual Display Dummy (Headless Ghost Adapter), manipulating the Guardian into thinking there is a monitor plugged in. The overall goal, however, was a success.



Figure 1 - Captured data from CPU

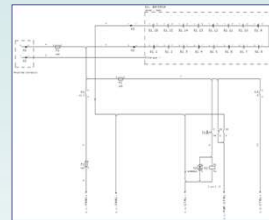


Figure 2 - Power Supply Diagram (Robotnik Guardian "Power and Control Manual")

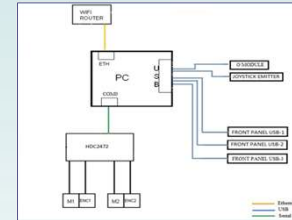


Figure 3 - Communication Diagram (Robotnik Guardian "Power and Control Manual")

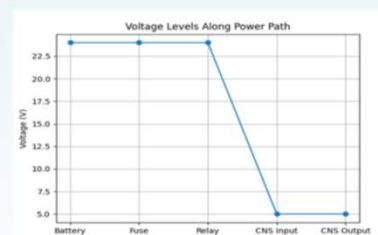


Figure 4 - Voltage Levels Along Power Path
Visual graph of level of voltage and pathway.

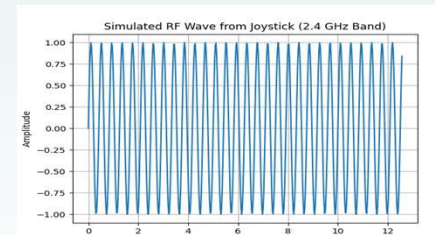


Figure 5 - Visualized simulation of signals from controller

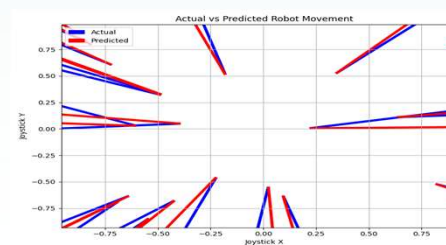


Figure 6 - Visualized prediction for future autonomy

Discussion

The work that was done ensured the knowledge of the functionality of the Robotnik Guardian. In the future, there are opportunities to proceed solving the issue with the Guardian robot.

Conclusion

By bridging electrical diagnostics with robotic software, this work ensured reliable system startup, validated safety-critical pathways, and enabled remote motion control. These efforts directly contribute to the robot's readiness for mentioned goals.

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